

Current Set-Up

- Pololu G2 18v15 x4
- Control 4 linear actuators

FA-240-S-12-2" (30:1)

- * 2.3 A current limit per actuator
 - ↳ Sometimes raised to 3A

Replacement Requirements

- 2.5 A (minimum)
- 11V - 14V (24V for no BEC)

Custom Board

Based off B00017

- Why no reverse polarity protection? ✓
 - ↳ XT30 connector only plugs in one way. Can't be accidentally flipped
- MCU selection considerations
 - ↳ Does MCU brand have to match motor driver ICs? X
 - ↳ Software preference?
 - ↳ Dev board vs. MCU IC ✓
 - Arduino Micro
- Actuators do not have feedback option ✓
- CAN vs. RS-485
 - ↳ Implementation

Initial Plan

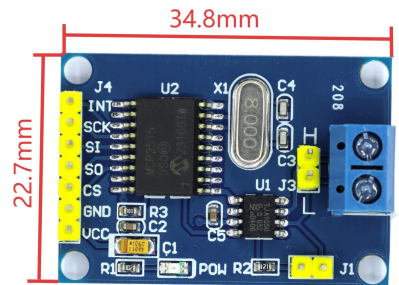
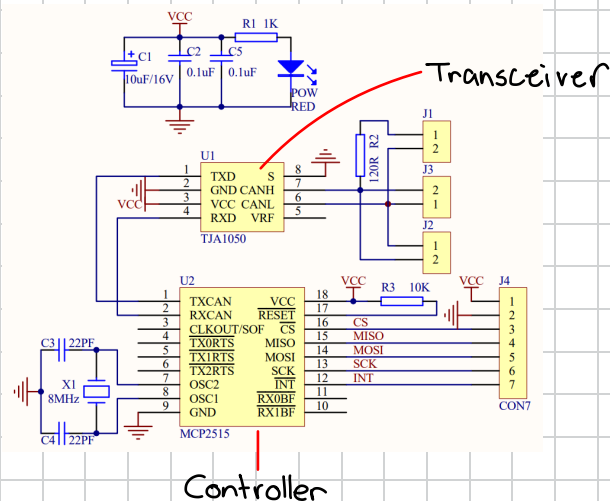
- Arduino Micro as MCU
 - ↳ 3 I/O pins + 2 A/D pins per $\frac{1}{4}$ -bridge
 - ↳ $V_{in} = 12V$
- IFX007T x8
- Voltage regulator IC ? **X**
 - ↳ Arduino and actuators both take 12V
- CAN communication
 - ↳ UART/I2C/SPI $\rightarrow$ CAN

Serial Comms

SPI $\rightarrow$ CAN converter

MCP2515

- Custom layout vs. premade module

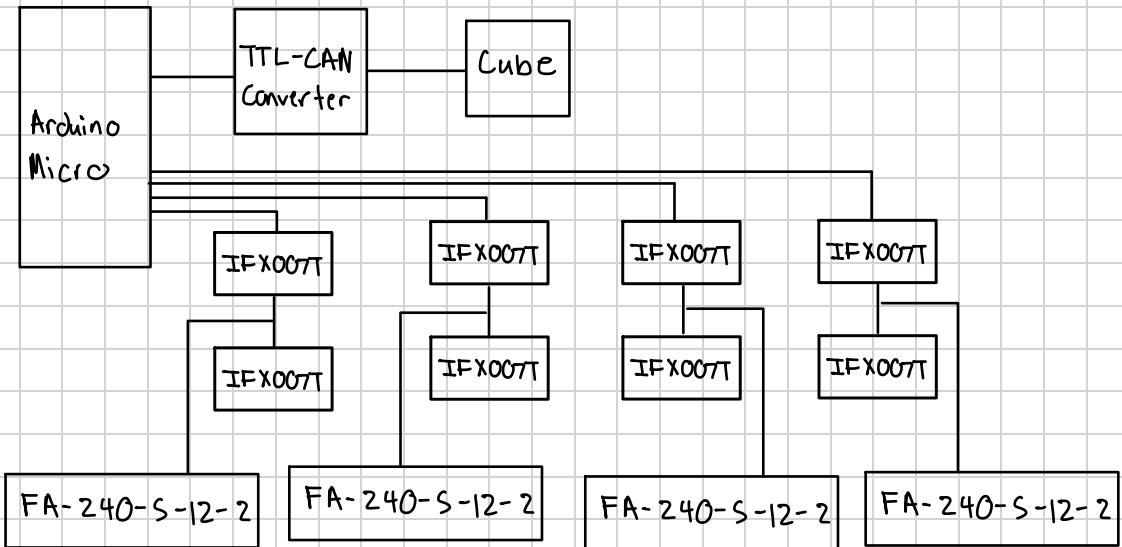


- Cube CAN port takes 4-pin JST-6H

Electronics List

- Arduino Micro
- WS TTL-CAN Converter
- Cube
- IFX007T x8
- IPD90P03P4L-04 x4

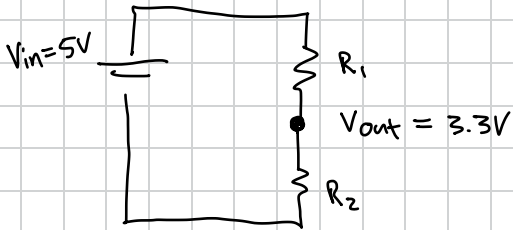
Basic



Voltage Divider

- WS-TTL-CAN takes 3.3V logic
- Arduino has 5V logic

Step down 5V $\rightarrow$ 3.3V for safety of CAN module



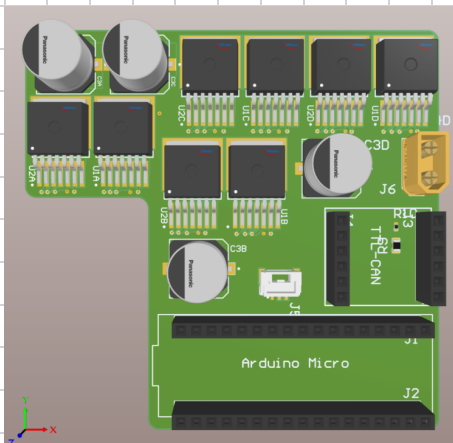
$$3.3/5.5 = 0.66$$

66% of resistance from R_2
34% of resistance from R_1

$$R_1 = 5V \quad R_2 = 10V$$

$$\frac{5V}{R} = 1 \times 10^{-3} A$$

Initial Layout

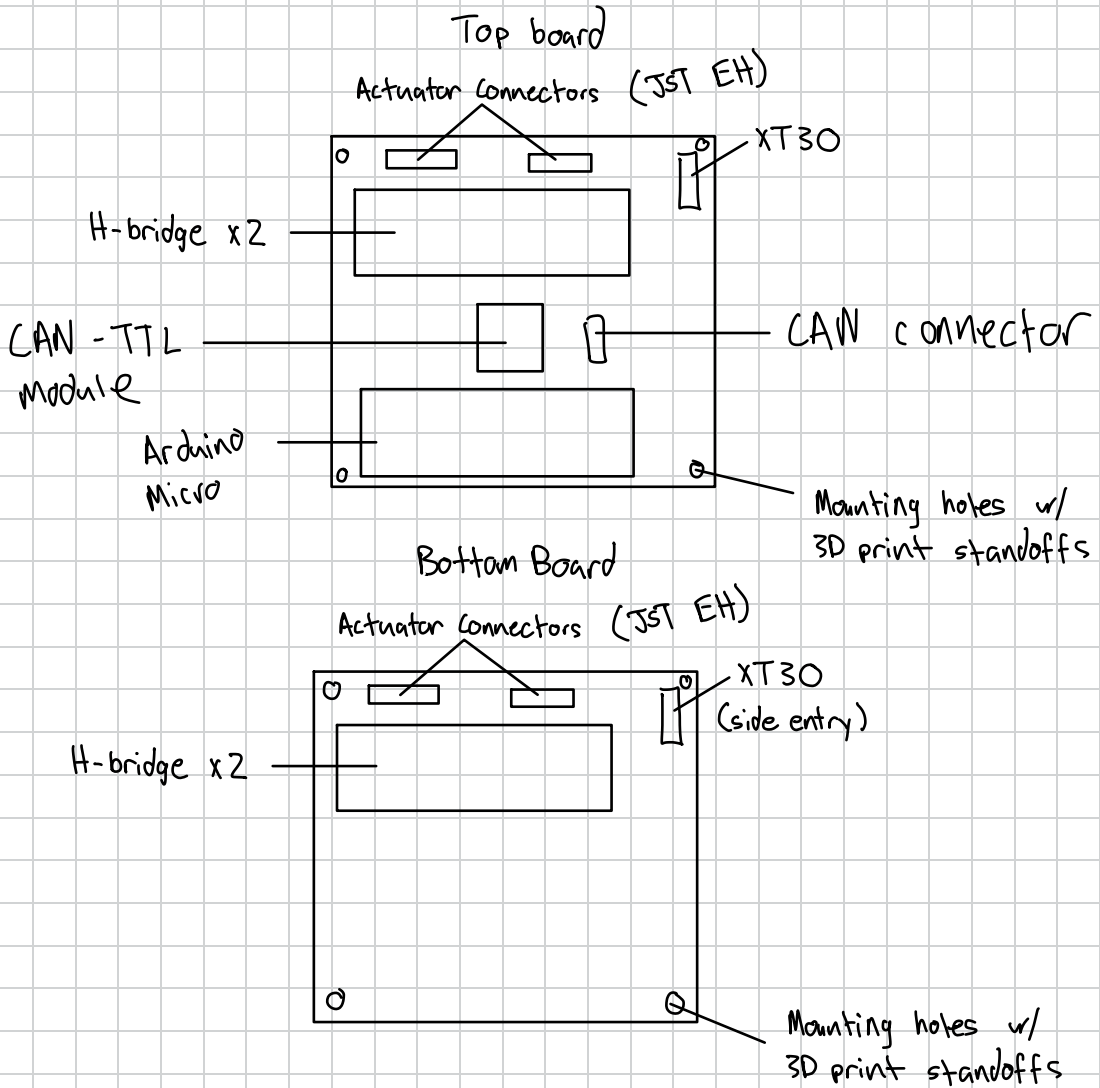


Problems:

- Caps on bottom
- Very tight space
- Footprint too big for available area on the robot

Revision Ideas

Stacked boards



* Adjust 7-in-1 board to take CAN instead of I²C

* Find female headers tall enough so caps fit

Revised Plan

- Teensy 4.0 MCU
- IFX007T Half-Bridge IC x8
- CAN Transceiver
- 24V - 3.3V regulator OR 24V - 5V regulator

July 6, 2026

- Replace 1000µF cap w/ smaller caps in parallel
- Connector for actuator output
↳ ideal wire gauge?
- CAN connector wiring

July 15, 2026

Board Concept Revision:

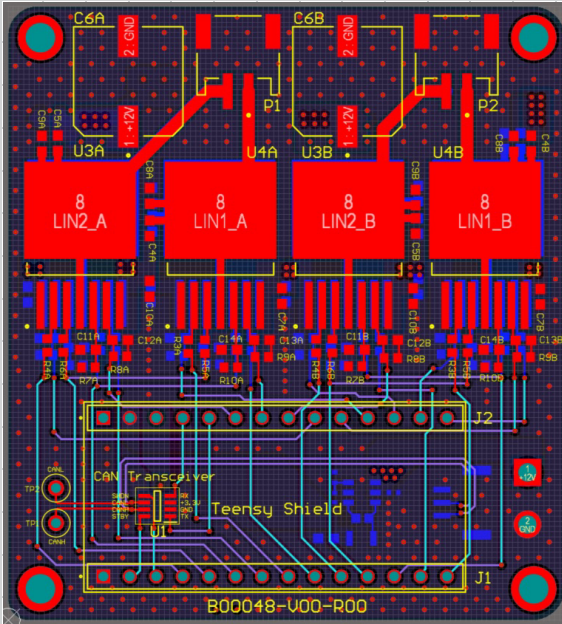
- Single board, no stacking
 - ↳ 4 half bridge ICs per side
- 8-layer board

Components + Analog signals
GND
+12V
Digital signals
GND
Digital signals
GND
Components + Analog signals

- Key considerations
 - ↳ Minimize switching noise from buck converter
 - ↳ Unobstructed GND return paths
 - ↳ Implement orthogonal routing

July 22, 2026

Current PCB Layout:



- 12V-5V regulation ✓
- CAN communication ✓
- Motor control/power ✓
- Actuator feedback ✗

Actuator Feedback Implementation

- Use V divider to step down V_{in} to logic HIGH voltage
↳ Connect to interrupt pin to detect falling edge i.e. power off
- Trigger memory write to store Hall sensor data after shutdown
- Bulk cap to slow down power off for MCU

July 23, 2026

Max board dimensions: 80mm x 54mm

Dev Board pin requirements:

- CAN RX and TX (2)
- Analog current sense pins (8)
- Digital GPIO (14 + 9)

$$\text{Total} = 8 \text{ Analog} + 23 \text{ Digital} + 2 \text{ CAN} = \boxed{33}$$

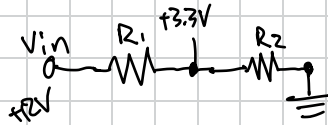
Teensy 4.1 fits *

Next steps:

- Find 2 row connector for new actuator output
- Ask SW team abt MCU preferences

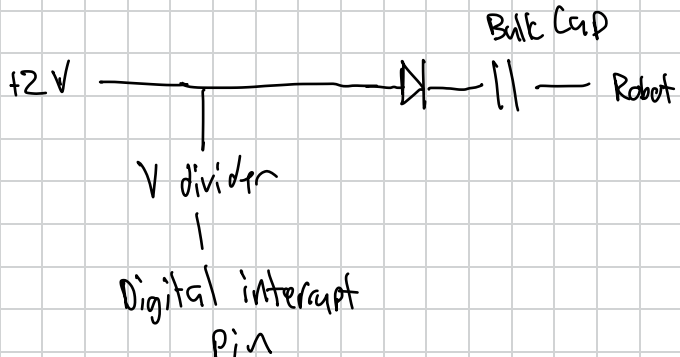
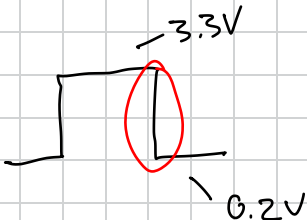
Voltage divider

$$\frac{3.3}{12} = 0.275$$



Digital pin recommended I draw

$$V = IR \quad 12V = (0.4 \times 10^{-3} A) R \quad R = 30000 \Omega$$



Total Resistance $40\,000\ \Omega$ for 3 mA digital pin input current.

$$R_1 = 29\,000\ \Omega \quad R_2 = 11\,000\ \Omega$$

15.85

38.15

DRG Violations

Solder Mask - Silver : 0.254

Silk - Solder mask : 0.1

Silk - Silk : 0.1

Design Review Feedback

1. Inductor selection

MCU draws 100 mA @ 600 MHz

buck must support $\approx 100\text{ mA}$ current draw

$$\text{let } I_{\text{out}} = 120\text{ mA}$$

$$V_{\text{in_max}} \approx 12\text{ V} \quad V_{\text{out}} = 5\text{ V} \quad K_{\text{ind}} = 0.5$$

$$L_{\text{min}} = \frac{12\text{ V} - 5\text{ V}}{120\text{ mA} \times 0.4} \times \frac{5\text{ V}}{12\text{ V} \times 2.1 \times 10^6} = 28.9\ \mu\text{H}$$

* Choose closest std value 27 or 33

$$\Delta i = \frac{V_{out}(V_{in_max} - V_{out})}{V_{in_max} \cdot L \cdot f_{sw}} = 48 \text{ mA}$$

(ripple I)

$$I_{peak} = 120 \text{ mA} + \frac{\Delta i}{2} = 0.144 \text{ A} \quad 144 \text{ mA}$$

* Use 1A output instead

$$\Delta i = 0.295 \text{ A}$$

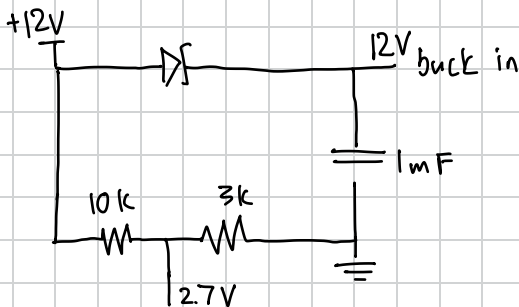
$$I_{peak} = 1.295 \text{ A}$$

Design Review 2 Changes

Teensy Current Draw: $\sim 100 \text{ mA}$

PA-10 Actuator: $10 \text{ mA} \times 2$

Total 5V rail $I_{load} = \sim 140 \text{ mA}$



Buck Converter Power

$$12V \times 0.85 \times I_{in} = 5V \times 0.15A$$

$$I_{in} = 0.073A = 73 \text{ mA}$$

Board Testing

- Power
 - ↳ Validate that 5V reg implementation works X
- CAN
 - ↳ Probe CAN to validate signal integrity
- Actuator drive ✓
 - ↳ Test control under dynamic loads ✓
 - ↳ Use I sense pin and PID control to manage current * Out of my scope
- EEPROM data write
 - ↳ Power shutdown test

GND 5V HALL HALL +12V -12V

Testing Log

First checking actuator polarity / direction

- $V_{\text{green-yellow}} = +12\text{V}$ extends
- $V_{\text{green-yellow}} = -12\text{V}$ retracts

Actuator homing when powered on

- Actuators should retract fully (0 position)
- Takes ~ 7.5 seconds to retract
 - ↳ Drive actuators w/ reverse polarity for 8.5 seconds
- IN1_A HIGH, IN2_A LOW should reverse

Board bring-up complete ✓

- 12 V 2A from PSU
 - ↳ 11.9V $V_{\text{G-12V}}$
 - ↳ 2.7V V_{sense}

Buck Converter Issues

No shorts between IC pins

- Checked for continuity between
 - ↳ 12V - GND, V_{in} - GND : MCU disconnected
 - ↳ 12V - GND, V_{in} - GND, 12V - buck out, V_{in} - buck out : MCU off

Noticing frequency foldback behaviour

- 133 kHz f_{sw} instead of 2.1 MHz
 - ↳ Triggered by low V_{in} , but 12V being supplied

Control loop not working

- FB at $\sim 2.3V$ not 1V it should be
- $V_{FB} > 1$ should trigger $\downarrow$ duty cycle
 - ↳ Behaviour is opposite, clamping to max duty. Why?

Both PCBs behave the same

- 98% duty cycle
- No matter what the FB pin reads

Continued Testing

Separate 12V and 5V rails

- Desolder diode
- Power MCU from separate PSU

Test firmware

- Actuators working ✓
- Check all 4 to validate pinouts

All 4 actuators working

- Pinout in Altium is correct ✓
- Needed to use PSU w/ higher current (5A $\rightarrow$ 10A)

Hall effect sensing

- A and D just reading 0. Why?
 - ↳ All connectors plugged into HALL 2
- When motors turned off, all started counting
- Plan: probe HALL signal with scope to see behaviour

Minor code changes

- Added volatile to all HALL-ISR variables
- Input → Input-pullup for GPIOs

Hall sensors working ✓

- Counts seem to correlate with Extension

2 of the actuators extend further than the others

- Swapped connectors to check why
 - ↳ Same actuators had the quirk ∴ not bc of PCB

Final Results

- Reliable actuator control (extension & retraction)
- Hall pulse tracking matches physical position
 - ↳ ↑ pulse count correlates to further extension
- Next steps include:
 - ↳ Validating CAN communication
 - ↳ Fixing/replacing buck converter
 - ↳ Testing power hold-up system